

E-Commerce Sales & Customer Analytics Task

Scenario:

You are working as a Junior Data Analyst for an Indian e-commerce startup.

The company wants insights from customer, product, and order datasets.

Datasets Provided:

1. Customers.csv :

https://drive.google.com/file/d/1cjztr8EteSGKQtP6de_8dUe7KM3zNOV7/view?usp=sharing

2. Products.csv :

https://drive.google.com/file/d/19gPTSBpAhNTL7n-Di7ONXIdbKXQJrQjY/view?usp=drive_link

3. Orders.csv :

https://drive.google.com/file/d/1Q9imhUU92nrWH-m-2tr9qGdNJplh2jz/view?usp=drive_link

TASKS

Part 1 :: Data Inspection & Cleaning

1. Load all datasets using Pandas.

2. Check:

- shape
- columns
- dtypes
- missing values
- duplicate rows

3. Clean the datasets:

- Handle missing values
- Remove duplicates
- Standardize text formatting

Part 2 :: NumPy Operations

1. Convert numeric columns into NumPy arrays.

2. Perform:

- mean
- median
- standard deviation
- min/maxNumPy

3. Demonstrate:
 - reshape
 - flatten
 - transpose
 - boolean masking
 - fancy indexing
4. Perform matrix multiplication example using NumPy.

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Part 3 :: Business Analysis

1. Find:
 - Top 5 customers by total spending
 - Most sold product
 - Highest revenue category
 - Average order quantity
 - City with most customers
2. Create new columns:
 - Total Price
 - Discounted Price
3. Perform:
 - Product-wise sales analysis
 - Customer-wise spending analysis
 - Monthly sales analysis

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Part 4 :: Filtering & Sorting

1. Filter:
 - Orders with discount > 10%
 - Delivered orders only
 - Customers from Ahmedabad
2. Sort:
 - Highest spending customers
 - Most expensive products

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Part 5 :: String & Date Operations

- 1. Convert customer names to uppercase.
 - 2. Extract:
 - month
 - year
 - weekday
- from order_date.

Expected Deliverables

- 1. Cleaned datasets
- 2. Python script / Jupyter notebook
- 3. Console-based insights
- 4. Properly commented code